ENVIRONMENTAL STUDIES

UNIT I & III

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Environmental Studies: Importance and Scope

What is Environmental Studies?

Environmental Studies (EVS) is the study of the environment and how humans interact with it.

Importance:

- Helps us understand **nature and ecosystems**
- Creates awareness about **pollution and climate change**
- Teaches how to **protect natural resources**
- Encourages **sustainable living**
- Helps in solving environmental problems

Scope of Environmental Studies:

- **Natural Resources** → water, forests, minerals
- **Ecosystems** → food chains, biodiversity
- **Biodiversity Conservation**
- **Pollution Control**
- **Social Issues** → global warming, waste management
- **Human Population & Environment**

Ecosystems

Concept of Ecosystem:

An ecosystem is a system where **living organisms interact with each other and their environment**.

Example: forest, pond, desert.

Structure of Ecosystem:

1. Biotic Components (Living)

- **Producers** → plants (make food)
- **Consumers** → animals (eat others)
- **Decomposers** → bacteria, fungi (break down waste)

2. Abiotic Components (Non-living)

- Sunlight
- Water
- Air
- Soil
- Temperature

Functions of Ecosystem:

- **Energy flow** (sun → plants → animals)
- **Nutrient cycling** (recycling of materials)
- **Food chains & food webs**
- **Ecological balance**

Pond Ecosystem:

Components:

- **Producers** → algae, aquatic plants
- **Consumers**
    - Primary → small fish, insects
    - Secondary → bigger fish
- **Decomposers** → bacteria

Features:

- Small and self-sustaining ecosystem
- Easy to study
- Shows clear food chain

Example Food Chain:
Algae → small fish → big fish → birds

Natural Resources and Management

Types of Natural Resources:

1. Renewable Resources

- Can be reused
- Examples: sunlight, wind, water

2. Non-Renewable Resources

- Limited supply
- Examples: coal, petroleum, minerals

Energy Resources:

Renewable Energy:

- Solar
- Wind
- Hydropower

Non-Renewable Energy:

- Coal
- Oil
- Natural gas

Water Conservation:

What is it?

Saving and managing water properly.

Methods:

- Fix leaks
- Use water efficiently
- Reuse water

Rainwater Harvesting:

Meaning:

Collecting and storing rainwater for future use.

Importance:

- Saves water
- Recharges groundwater
- Reduces water shortage

Watershed Management:

Meaning:

Management of land and water in a specific area (watershed).

Importance:

- Prevents soil erosion
- Improves water availability
- Supports agriculture
- Maintains ecological balance

Questions and Answers:

2 Marks Questions:

1. What is natural ecosystem? Give an example.

A natural ecosystem is a system where living organisms interact with each other and the natural environment without human interference.
Example: Forest ecosystem, pond ecosystem.

2. Define rainwater harvesting. Mention it's any two types.

Rainwater harvesting is the process of collecting and storing rainwater for future use.
Types:

- Rooftop rainwater harvesting
- Surface runoff harvesting

3. Mention the two control measures of water pollution.

- Proper treatment of industrial and sewage waste before discharge
- Reducing use of harmful chemicals and plastics

4. What is the meaning of global warming? Mention any one cause.

Global warming is the increase in Earth's average temperature due to greenhouse gases.
Cause: Burning of fossil fuels like coal and petrol.

5 Marks Questions:

1. Explain non-renewable energy resources with example.

Non-renewable energy resources are resources that cannot be replaced quickly once they are used. They are limited in quantity and take millions of years to form.

Examples:

Coal, Petroleum, Natural gas

Features:

- Exhaustible in nature
- High energy output
- Cause environmental pollution

Conclusion:

Non-renewable resources are important but should be used carefully and replaced with renewable energy sources.

2. Give an account on plastic pollution and its impact.

Plastic pollution is the accumulation of plastic waste in the environment.

Causes:

- Excess use of plastic products
- Improper disposal

Impacts:

- Harms animals and marine life
- Causes soil and water pollution
- Blocks drainage systems

Conclusion:

Reducing plastic use and proper recycling can help control plastic pollution.

3. Discuss the effects of ozone layer depletion and mention its preventive measures.

The ozone layer protects Earth from harmful UV rays.

Effects:

- Skin cancer and eye damage
- Damage to plants and crops
- Global warming increase

Preventive Measures:

- Reduce use of CFC gases
- Use eco-friendly products
- Promote environmental awareness

4. Write a note on Solid Waste Management.

Solid waste management is the process of collecting, treating, and disposing of solid waste properly.

Steps:

- Collection
- Segregation
- Recycling
- Disposal

Importance:

- Reduces pollution
- Protects health
- Saves resources

5. Give an account of sustainable development goals.

Sustainable Development Goals (SDGs) are global goals set to protect the environment and improve quality of life.

Key Goals:

- Clean water and sanitation
- Climate action
- Affordable and clean energy

Importance:

- Promotes balanced development
- Protects environment for future generations